MILITARY SPECIFICATION

AMPLIFIER-CONTROL GROUP, AN/SIA-122(V)

1. SCOPE

1.1 This specification covers the detailed requirements for an amplifier voice intercommunication system for transmission of two-way communications between various aviation ordnance and missile handling stations in an Aircraft Carrier.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-62 - Capacitors, Fixed, Electrolytic DC Aluminum, Dry Electrolytic, Polarized.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements For.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-3965 - Capacitors, Fixed, Electrolytic Tantalum.
MIL-L-12606 - Loudspeaker, Permanent Magnet Unencased 4-Inch Diameter Cone, 2-Watt, Gunblast and Immersion-Resistant, Type LS-445/V.
MIL-A-21577 - Amplifiers Audio Frequency; and Amplifier Control Groups --- Shipboard Announcing.

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Preferred and Guidance Lists of Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

FSC 5830

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3. REQUIREMENTS

3.1 The amplifier-control group AN/SIA-122(V) shall be in accordance with the applicable paragraphs of MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.2 Amplifier-control group.- The system complement shall consist of on amplifier control assembly, two 30 station control units and control, speaker-talk stations with explosion proof loudspeakers. The quantity of speakers and control, speaker-talk stations shall be as specified in the contract or order. Requirements for the explosion proof loudspeakers is not covered in this specification.

3.3 Amplifier control assembly.- The amplifier control assembly is the basic equipment in the system. The assembly shall provide for two independent amplified audio channels to which up to thirty speaker-talk-back stations per channel may be operated. The assembly shall also provide "audio override" by the speaker-talk-back stations.

3.3.1 Enclosure.- The enclosure shall be of formed sheet metal constructed with internal supporting frame work. The enclosure shall be designed for bulkhead mounting. The enclosure shall be dripproof protected from the top. Force air cooling shall not be permitted. Amplifier modules shall be mounted in a vertical position to facilitate the unrestricted flow of air.

3.3.2 Maximum dimensions and weight.- The dimensions shall not exceed 16 inches in height, 14 inches in width and 14 inches in depth. Weight shall not exceed 100 pounds.

3.3.3 Controls.- The following controls and components shall be provided on the front of the enclosure:

(a) Two 20 watt plug-in type power amplifiers with individual power supplies, one for each audio amplifier channel.
(b) Two amplifier power supply switches, one for each audio amplifier channel, connects ship power supply originating from thirty station control units to amplifiers.
(c) Two relay power supply switches, one for each audio channel, connects ships power supply originating from thirty station control units to relay power supplies.
(d) Two power available indicator lamps, indicates when the to power supplies are connected to amplifier control from the thirty station control units.

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(e) One amplifier output test meter, indicates output level of amplifier channel under test.
(f) Two test switches, connects to test meter output and load resistor to channel under test.
(g) Two microphone receptacles, connectsan MPM-6 microphone to amplifier for test purposes.
(h) Eight fuses for relay power and amplifier power supply protection.

Relays, wiring, terminals and other components as required shall be provided within the enclosure.

3.3.4 Relays.- Relays except power type contactors may be hermetically sealed.

3.3.5 Alarm oscillator.- Alarm oscillation will not be required.

3.3.6 Power amplifier modules.- The power amplifier modules shall be constructed with quick disconnect features designed to permit easy removal of the modules from the amplifier control for servicing. The amplifier modules shall be provided with individual gain controls, accessible through the front panel.

3.3.6.1 Output power.- The rated output power for the individual power amplifier modules shall be 20 watts, minimum rating.

3.3.6.2 Input impedance.- Input impedance for the individual power amplifier modules shall be 150 ohms (plus 10 minus 0 ohms) resistive at 1000 cycles per second (cps).

3.3.6.3 Frequency response.- Frequency response of the power amplifiers shall be measured as specified in MIL-A-21577, except that the amplifiers shall be delivering 10 db. below rated output at 1000 cps.

3.3.6.4 Distortion.- Distortion of the power amplifiers shall be measured as specified in MIL-A-21577, except that at 200 and 6000 cycles, the measurement shall be made at 6 db. below rated output.

3.3.6.5 Semiconductor devices.- Semiconductor devices shall be employed in the design of the power amplifiers and shall be selected from MIL-STD-701.

3.3.6.6 Capacitors.- Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in pre-amplifiers and power amplifiers. Plug-in type electrolytics in accordance with MIL-C-62 may be utilized in the power supply.

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3.3.6.7 Wiring.- Wiring employed shall have type T insulation as shown in the applicable table of MIL-I-983. Printed wiring, when employed, shall be in accordance with type C lead termination specified in MIL-STD-275.

3.3.6.8 Interstage input circuit.- Input impedance and input level control will not be required.

3.3.7 Accessories.- A six foot test cable shall be provided for testing power amplifier modules when removed from the amplifier control.

3.4 Thirty station control assembly.- The two thirty station control assemblies shall be identical in construction and operation and each assembly shall provide facilities for selecting and connecting any number of the thirty speaker-talk-back stations to one channel of the amplifier control assembly. Operation of any of the station selector switches, followed by operation of the speaker-talk-listen switch shall operate the necessary relays in the amplifier control assembly to provide audio communication to the selected speaker-talk-back stations. In addition to the audio communication, operation of the selector switch shall provide a busy indication at all thirty speaker-talk-back stations. Release of the speakers' talk-listen switch shall allow hands-free reply from speaker-talk-back station selected.

3.4.1 Enclosure.- The enclosure shall be formed sheet metal, designed for desk mounting. The enclosure shall be dripproof from the top.

3.4.2 Maximum dimensions and weight.- The dimensions shall not exceed 10 inches in height, 16 inches in width, and 8 inches in depth. Weight shall not exceed 25 pounds.

3.4.3 Controls.- The following controls shall be provided on the front of the enclosure:

(a) Power supply switch, connects ship's power supply to one channel of the amplifier control assembly.
(b) Power available indicator, indicates when ship's power supply is available.
(c) Thirty talk-back station selector switches, connects remote speakers-talk-back station to amplifier input.
(d) Loudspeaker (reproducer microphone).
(e) Volume control, controls level of incoming audio from the talk-back station.
(f) Press-to-talk switch, connects loudspeaker to amplifier input when used as a microphone.
(g) Two fuses, protects amplifier control assembly power supply.
(h) Busy indicator, indicates when system is in use.

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3.4.4 Loudspeaker.- The loudspeakers shall be type LS-445 in accordance with MIL-L-12606.

3.5 Control, speaker-talk-station.- The control unit shall provide facilities for initiating emergency reports to its respective thirty station control unit. The "Emergency-Report" switch shall be provided speech priority over the thirty station control unit. In addition, operation of this switch shall provide busy indications at all thirty speaker-talk-back stations and the thirty station control unit.

3.5.1 Enclosure.- The enclosure shall be designed for bulkhead mounting and shall be explosionproof in accordance with MIL-E-2036. The enclosure shall be dripproof.

3.5.2 Maximum dimensions and weight.- The dimensions shall not exceed 6 inches in height, 6 inches in width and 5 inches in depth. Weight shall not exceed 10 pounds.

3.5.3 Controls.- The following controls shall be provided on the front of the enclosure:

(a) Busy indicator lamp, indicates when system is in use.
(b) Emergency report switch, allows operator to report and override outgoing reports from the thirty station control unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling, examination and tests shall be as specified in MIL-A-21577, in addition the "control, speaker-talk stations" shall also be tested as specified in MIL-E-2036.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-21577.

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6. NOTES

6.1 The notes specified in MIL-A-21577 are applicable to this specification.

Preparing activity:
Navy - Ships
(Project 5830-N059Sh)

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